

User Churn Project | Exploratory Data Analysis

Prepared for: Waze Leadership Team

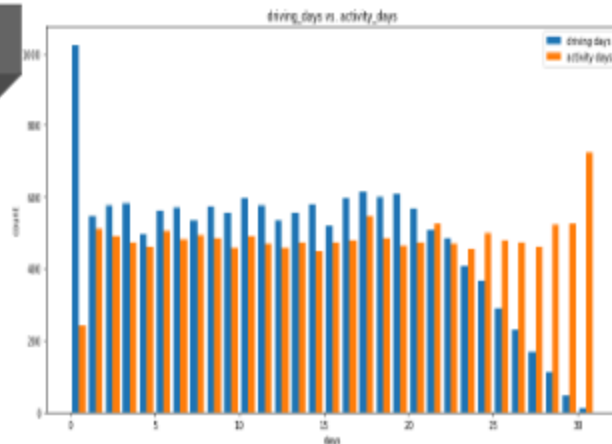
Project Overview

To accelerate platform growth, the Waze data team is leveraging exploratory data analysis (EDA) to understand and prevent monthly user churn. Pinpointing early churn indicators allows for targeted, proactive retention campaigns that protect customer lifetime value. This report summarizes the key insights from Milestone 3, which will directly shape our upcoming predictive modeling and development roadmap.

Details

Key Insights

- High app engagement prevents churn. Users driving all 30 days had a 0% churn rate, compared to a 40% churn rate for inactive users. Frequent driving days over the month strongly align with higher retention.
- A positive correlation exists between daily driving distance and churn, the farther a user travels per driving day, the more likely they are to churn.
- User representation is stable and evenly spread across all tenure lengths up to roughly 10 years.
- Features are primarily right-skewed (concentrated at the low end) or uniformly distributed (equally distributed). However, extreme, potentially anomalous outliers are present in `driven_km_drives`, `activity_days`, and `driving_days`.



The churn rate is highest for people who didn't use Waze much during the last month.

Next Steps

- We need to investigate why long-term users exhibited a sudden, sharp spike in app usage over the last month
- we confirm all monthly metrics share the same logging period? (Note: Some variables max out at 30 days, others at 31).